

Erica Holdmore, Ph.D.

 Auburn, MA
 [linkedin.com/in/erica-holdmore-49544580/](https://www.linkedin.com/in/erica-holdmore-49544580/)

 (818) 850-8805

 eholdridge@gmail.com
 github.com/eholdmore

PROFESSIONAL SUMMARY

Technical Product Lead & Computational Biologist with experience leading the end-to-end lifecycle of AI-driven healthcare products. Expert in moving high-impact products from concept to production-ready assets. Proven track record in orchestrating cross-functional groups of engineers, data scientists, and subject area experts to deliver scalable platforms integrating EHR, genomic, and real-world data.

EDUCATION

Ph.D.
Ecology & Evolutionary Biology
Yale University

M.S.
Biology
California State University

B.S.
Biology
Florida State University

EXPERIENCE

Project Lead, Computational Biologist III - Dana-Farber Cancer Institute 01/2023 – Present

- Lead the roadmap, backlog prioritization, and milestone planning for AI-enabled clinical research platforms supporting translational oncology across multiple disease centers.
- Partner with clinical leadership, physicians, and institutional executives to translate complex clinical workflows into technical specifications and deliverable product features.
- Design scalable architectures integrating EHR data, ICD-10/OncoTree ontologies, genomic sequencing, and NLP-derived insights from clinical notes and radiology reports.
- Direct internal computational scientists and external contractors in an Agile environment to build production-ready data pipelines for cohort discovery and trial feasibility.
- Oversee HIPAA-compliant system security reviews and improved genomic analysis pipelines for 50,000+ tumor profiles by implementing statistical quality control frameworks.

Assistant Professor of Bioinformatics and Biology - Stonehill College 09/2022 – 01/2023

Established and led a grant-funded computational biology program, managing start-up funds to launch research initiatives and develop advanced bioinformatics curricula.

Postdoctoral Research Fellow - Boise State University Biological Sciences 01/2021 – 09/2022

Secured \$207K in federal funding to architect end-to-end metagenomic pipelines using Python and HPC environments for the study of complex microbial communities.

TECHNICAL SKILLS

Product & Agile Leadership: Roadmap Planning, Backlog Prioritization, User Stories, Stakeholder Engagement, Scrum, Jira/Confluence, GitLab

Clinical Informatics & AI: Data Harmonization, NLP/LLM Integration, HIPAA-Compliant Architecture, Clinical Ontologies

Engineering & Data Science: Python, R, SQL, Streamlit/Reflex, Statistical Modeling, Data Visualization, Evidence-Based Narratives

Bioinformatics & Infrastructure: Pipeline Development (WDL/Nextflow), Docker/Singularity, Cloud (GCP/Terra), HPC Environments

HOW I MANAGE PRODUCTS

Clinical Trial Portfolio Management Platform

Dana-Farber Cancer Institute | Project Lead

Scalable platform for clinical trial portfolio management, search, and analytics across disease centers.

THE CHALLENGE: This project originated as a rapidly developed Streamlit + SQLite prototype designed to test and refine clinical trial management workflows with end users. After early adoption demonstrated clear institutional value, I was tasked with leading the product's evolution from proof-of-concept to production system by redesigning the platform around a scalable Reflex + PostgreSQL architecture that could support enterprise-scale growth and ongoing feature development.

TECHNICAL ARCHITECTURE

Technology Stack:

Frontend: Reflex
Backend: Python application services
Database: PostgreSQL
Infrastructure: Docker
CI/CD: GitLab Pipelines

Data Integration: Engineered a scalable relational schema to harmonize clinical trial metadata, patient enrollments, and disease center trial curation.

Design Principles: Focused on system scalability and feature parity. By implementing a reproducible development workflow, I lowered the technical barrier for new contributors, ensuring engineer onboarding in under 30 minutes.

FOUR-MONTH DEVELOPMENT TIMELINE

Kick-Off	Architecture	Implementation	Testing	Launch
<i>Environment set-up, feature parity checklist, initial risk assessment</i>	<i>Database schema, CI pipeline, deployment strategy</i>	<i>App staged, vertical slice architecture, end-to-end workflows</i>	<i>Parity with prototype, risk-based testing, UI smoke test</i>	<i>User acceptance testing, production deploy, 2-week hypercare</i>

LEADERSHIP

AI-Assisted Workflow: Established AI-assisted development workflows leveraging Claude Code for specification generation, rapid prototyping, and implementation planning.

Stakeholder Alignment: Established meeting cadence (15-minute progress checks, bi-weekly stakeholder updates) to maintain institutional alignment while protecting engineering focus.

Technical Documentation: Built a centralized technical repository in Confluence and Loom, encompassing PRDs, system architecture documentation, and video-based UI walkthroughs.

IMPACT & KEY METRICS

Reduced time-to-insight for clinical trial discovery by centralizing disparate data streams, enabling multi-center expansion without further architectural overhauls.

- Manages 150+ clinical trials across three major disease centers.
- Supports 100+ monthly active users, including research nurses, clinicians, and principal investigators.
- Improved performance and reliability through a modular application structure, supporting future institute-wide expansion without further architectural overhauls.